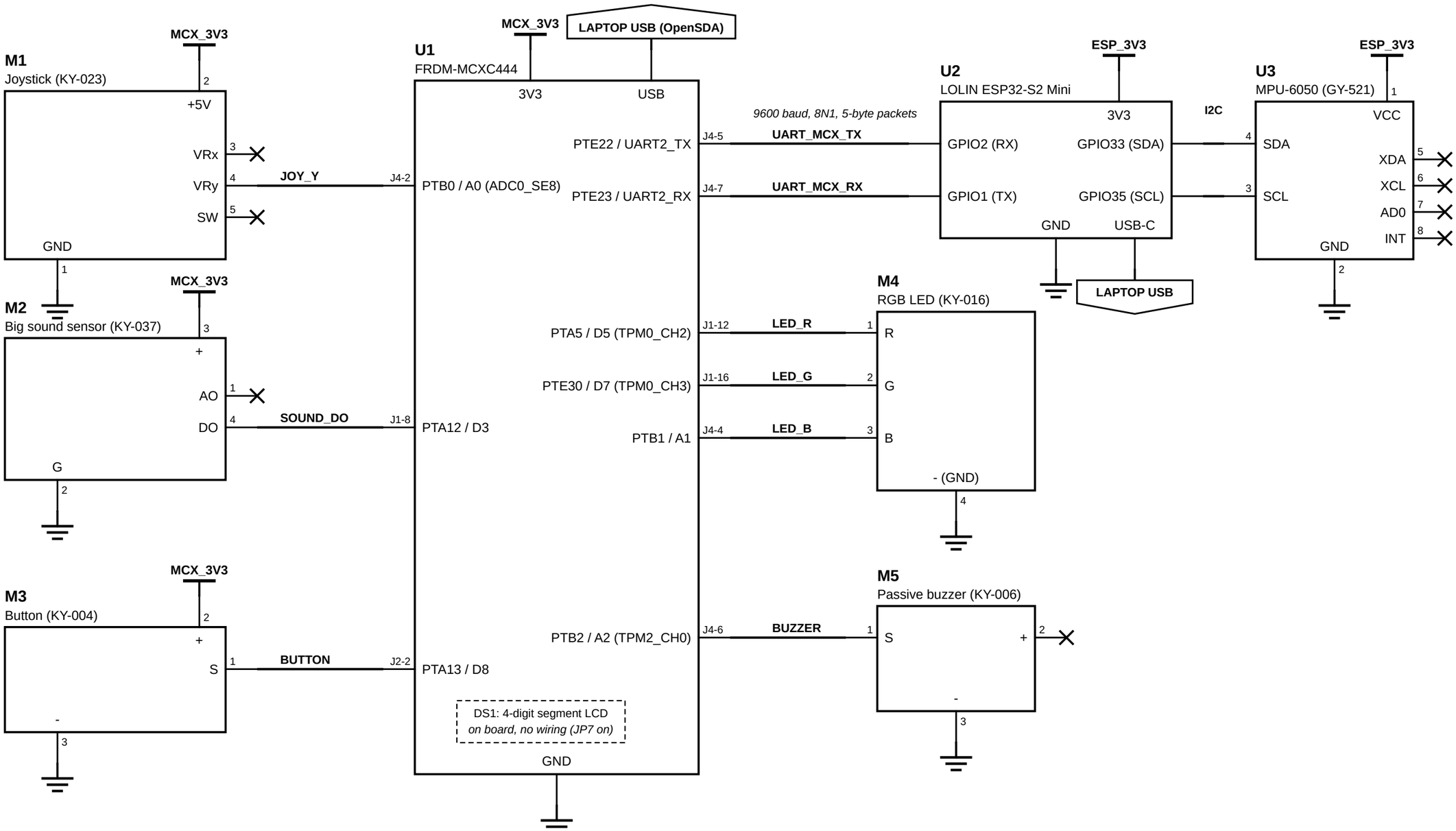

NOTES

1. Power every kit module from 3.3 V (MCX_3V3 = FRDM 3V3 pin, ESP_3V3 = S2 Mini 3V3 pin). Do not join the two 3V3 rails.
2. All GND symbols are one net: run one wire from a FRDM GND pin to an S2 Mini GND pin (common ground for the UART).
3. M1 pin "+5V" is only the module's silkscreen name: wire it to 3V3 so the joystick output stays inside the ADC range.
4. UART is crossed: MCX TX (PTE22) -> ESP RX (GPIO2), MCX RX (PTE23) <- ESP TX (GPIO1). Both sides are 3.3 V logic.
5. Small labels on U1 pins are FRDM header positions (e.g. J4-2). No external resistors: the kit modules carry their own.
6. X = not connected. MPU-6050 AD0 left open (I2C address 0x68). GY-521 has on-board I2C pull-ups.
7. Both USB ports go to the laptop: OpenSDA powers and flashes the FRDM; the S2 Mini USB-C powers it and acts as the keyboard.

Title	**Doomscroll Breaker - System Wiring**
Project	CG2271 Mini Project FRDM-MCXC444 + LOLIN ESP32-S2 Mini + sensor kit
Size: A3	Sheet 1 of 1
Date: 20/10/2026	Rev: 2.0 (B/W)